MANDAAR RAO

mandaar.rao1@gmail.com | LinkedIn | GitHub | Portfolio

EXPERIENCE

Tata Elxsi Limited | *Cyber Security Engineer* | Feb 2025 – Sep 2025

- Applied STRIDE and TARA threat modeling across cloud, web, and mobile platforms for a smart mobility ecosystem, documenting 3,000+ attack paths across northbound, southbound, and east/west interfaces. Delivered a risk-prioritized mitigation roadmap to strengthen ISO/SAE 21434 compliance.
- Developed a Cyber Resilience Act (CRA) compliance checklist adopted by 3 internal product teams and EU clients, aligning connected-product designs with CE certification standards and streamlining regulatory readiness.
- Identified and remediated critical injection, broken authentication, and API security vulnerabilities during VAPT of web and GenAI applications, reducing exploitable attack surface across the SDLC.
- Delivered VAPT and threat modeling workshops (STRIDE, TARA) to 500+ incoming graduate engineers across multiple hiring batches, improving security awareness and reducing insecure code patterns in post-training reviews.

SecureThings.ai Pvt. Ltd. | *Associate Security Engineer* | Jul 2023 – Oct 2024

- Conducted full-scope web application penetration testing on an automotive cybersecurity SaaS platform in AWS, uncovering critical OWASP Top 10 vulnerabilities (SQLi, IDOR, broken authentication) and hardening the overall attack surface.
- Built an Android pen-test environment using Frida and Burp Suite for a connected-vehicle app, bypassed SSL pinning and root detection, and surfaced sensitive data exposure and insecure API communication risks before production release.
- Developed a dark web OSINT scraper monitoring 20+ threat actor forums for automotive security incidents, accelerating emerging exploit detection by 70% and enabling proactive incident response.
- Built an Automotive CVE catalog mapping 200+ CVEs to MITRE ATT&CK and CVSS, improving risk prioritization and reducing mean-time-to-triage for the vulnerability management team.
- Engineered a FastAPI-based Splunk-MongoDB integration pipeline for a vehicular SOC, enabling real-time log ingestion from 50+ vehicle endpoints and improving threat detection coverage.

PROJECTS

Recon-Guru | Automated Attack Surface Recon Framework | GitHub | 2023–2025

- A modular Python recon framework chaining 8 OSINT sources, async subdomain probing, tech fingerprinting, full-port scanning, NVD CVE/ExploitDB lookup, and Shodan enrichment; outputs structured JSON and HTML reports with stealth mode and Docker support.

Zero Trust Adaptive Security Gateway for IoT | GitHub | 2025

- Trained a Random Forest classifier on the IEEE IoT Network Intrusion Dataset to generate per-flow probabilistic threat/trust scores; auto-derived device- and protocol-specific Zero Trust access policies from high-trust traffic baselines and translated them into deployable Snort IDS rules for anomaly detection.

Finmo | File Integrity Monitor | GitHub | 2026

- Cross-platform (Linux/Windows) Python FIM tool using SHA-256 hashing and SQLite baselines to detect unauthorized file changes in real time via watchdog, with multi-channel alerting (Telegram, Gmail) and YAML-driven config for watched paths and exclusions.

SKILLS

Cyber Security: Penetration Testing (Web, API, Android), Threat Modeling (STRIDE/TARA), VAPT, Red Team TTPs, Application Security, Cloud Security (AWS), Zero Trust Architecture, IDS/IPS, SIEM Operations, Incident Response, Secure SDLC, Vulnerability Management, Threat Intelligence, Secure Code Review, Cryptography

Programming & Scripting: Python, Bash, SQL, Java, C++

Tools: Burp Suite, Metasploit, Nmap, Wireshark, Frida, Ghidra, Splunk, Hashcat, Nessus, Elastic SIEM, Sysmon, MongoDB, FastAPI, Git, AWS, Docker

Frameworks & Standards: MITRE ATT&CK, OWASP Top 10, CVSS, NIST NVD, ISO/SAE 21434, EU Cyber Resilience Act (CRA)

CERTIFICATIONS

- CompTIA Security+ | Credential
- CompTIA CySA+ | In Progress
- Google Cybersecurity Certificate | Credential
- ISO/IEC 27001:2022 Information Security Associate | Credential

PUBLICATION

- **IEEE Xplore** - A Review of the OWASP Top 10 Web Application Security Risks and Best Practices for Mitigating These Risks: Analyzed real-world breaches (Target, Equifax) using OWASP Top 10 to evaluate security shortcomings and proposed practical mitigation strategies.

EDUCATION

University of Washington, Bothell | M.S. *Cyber Security Engineering* | Sep 2025 – May 2027 | GPA: 3.85/4.0

- Coursework: Cryptography, Information Assurance, Network & System Security, Malware Analysis & Reverse Engineering

Savitribai Phule Pune University | B.E. *Information Technology* | Aug 2019 – Apr 2023 | GPA: 3.31/4.0

- Coursework: Computer Networks & Security, Operating Systems, AI/ML, Distributed Systems